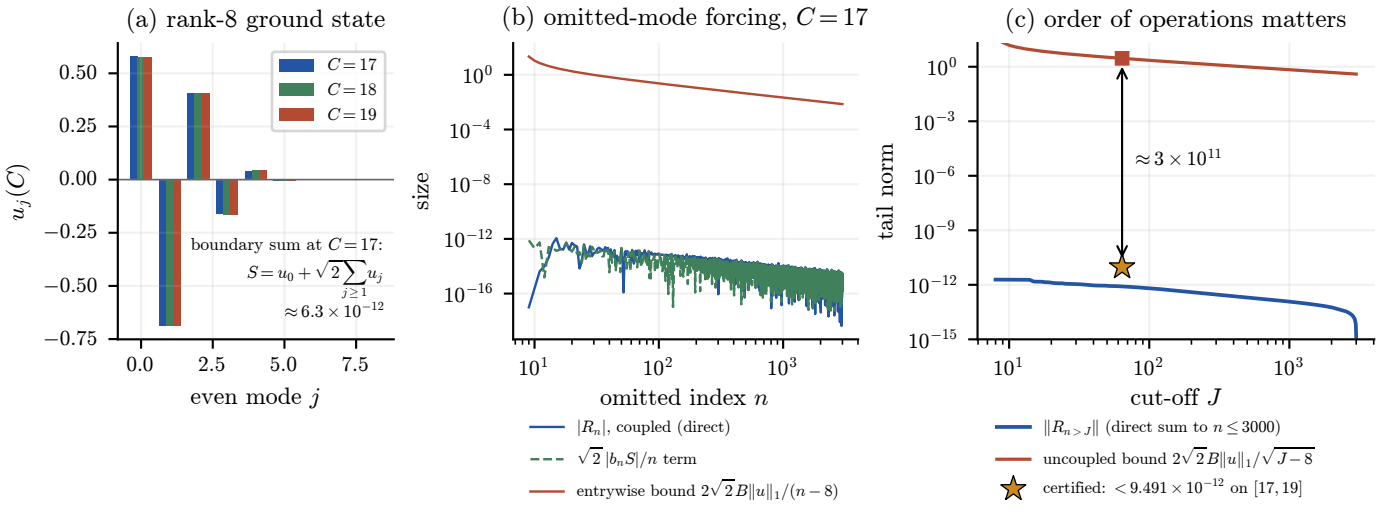
(a) rank-8 ground state
$u_j(C)$
0.50
0.25
0.00
−0.25
−0.50
−0.75
$C = 17$
$C = 18$
$C = 19$
boundary sum at $C = 17$:
$S = u_0 + \sqrt{2}\sum_{j\geq 1} u_j$
$\approx 6.3 \times 10^{-12}$
0.0
2.5
5.0
7.5
even mode j
(b) omitted-mode forcing, $C = 17$
size
10^{0}
10^{-4}
10^{-8}
10^{-12}
10^{-16}
10^{1}
10^{2}
10^{3}
omitted index n
$|R_n|$, coupled (direct)
$\sqrt{2}\,|b_n \mathcal{S}|/n$ term
entrywise bound $2\sqrt{2}B\|u\|_1/(n-8)$
(c) order of operations matters
tail norm
10^{0}
10^{-3}
10^{-6}
10^{-9}
10^{-12}
10^{-15}
$\approx 3 \times 10^{11}$
10^{1}
10^{2}
10^{3}
cut-off J
$\|R_{n>J}\|$ (direct sum to $n \leq 3000$)
uncoupled bound $2\sqrt{2}B\|u\|_1/\sqrt{J-8}$
certified: $< 9.491 \times 10^{-12}$ on $[17, 19]$